

Paths & Connectivity

Walks $\rightarrow$ non empty list , each edge connects the vertex before it and the vertex after it.

The **length of a walk** means the number of edges used.

Same vertex ke multiple times pass kora jay walk a

Trail : Walk - repeated edges

Path : Walk - repeated Vertex

Circuit : Train with two same end vertices

Closed path : Path with the same edged vertices

Walk length $\rightarrow$ The number of edges

Maximal Connected Subgraph : A connected part of a graph that cannot be made bigger without losing the idea that it is one connected piece.

Cut Edge : An edge whose deletion increases the number of components if and only if it belongs to no cycle

Every u,v -walk contains a u,v -path.

Every graph with n vertices and m edges has at least $n-m$ connected components

Let G be a simple graph of n vertices. If G has exactly k components, then the number m of edges of G satisfies
$$n - k \leq m \leq (n-k)(n-k+1)/2$$

A graph G is Bipartite if and only if it contains no odd cycles, meaning its vertices can be partitioned into two disjoint sets where every edge connects a vertex in one set to the other.

A connected graph G is Eulerian if and only if every vertex of G has even degree.

A connected graph G is Eulerian if and only if G has a cycle decomposition.

Let G be a simple Hamiltonian graph. Then for each subset S of $V(G)$, the number of connected components in $G - S$ is at most $|S|$

Let G be a simple graph of n vertices. Let u and v be two vertices in G such that $(u,v) \notin E(G)$ and $d_G(u) + d_G(v) \geq n$. Then G is Hamiltonian if and only if $G + (u,v)$ is Hamiltonian.

Let G be a simple graph of $n \geq 3$ vertices. Then G is Hamiltonian if $d_G(u) + d_G(v) \geq n$ for every pair of non adjacent vertices u and v in G .

Trees

Tree

Leaf of tree

Internal Vertex

Forest (acyclic graph)

Maximal Path : a **maximal path** is a path that cannot be extended any further by adding another edge to either of its ends without breaking a specific rule.

Rooted Tree

k-array tree

Every Vertex except the root is a Child Vertex

Binary Tree

An Ordered rooted tree

A **labeled tree** is a tree where every single vertex is given a unique identifier

Every tree with two or more vertices has at least two leaves.

Every edge in a tree is a cut edge.

Let G be a graph with n vertices. Then, any two of the following statements imply the third (and characterize a tree of n vertices).

(a) G is connected.

(b) G contains no cycle.

(c) G has $n - 1$ edges.

(1) A forest with n vertices and k components has $n - k$ edges.

(2) A forest with n vertices and m edges contains $n - m$ components.

(3) A graph with n vertices, m edges and $n - m$ components is acyclic.

A simple graph G is a tree if and only if for each pair of vertices u and v of G , there is a unique u, v -path in G

Every connected graph contains a spanning tree

Let $G = (V, E)$ be a connected graph and let $S \subseteq E$ such that $G - S$ is disconnected. Then every spanning tree of G has an edge in S .

Let G be a connected graph and let e be an edge in G . Then $\tau(G) = \tau(G - e) + \tau(G \setminus e)$.

There are n^{n-2} distinct labeled trees of n vertices.

PigeonHole Principle

Planar Graph

a planar graph can have more than one planar embedding.

The embedding of a 2-connected planar graph G is unique if and only if G is a subdivision of a 3-connected graph /
every 3-connected planar graph has a unique plane embedding.

Let G be a connected plane graph, and let n , m , and f denote, respectively, the numbers of vertices, edges, and faces of G . Then $n - m + f = 2$

If G is a planar graph with $n(\geq 3)$ vertices and m edges, then $m \leq 3n - 6$. Moreover the equality holds if G is maximal planar.

Let G be a planar graph. Then G has a vertex of degree at most 5.

Every maximal planar graph of four or more vertices has at least four vertices of degree five or less.

Show that Petersen graph is nonplanar. (Use Kuratowsk's Theorem.)

Show that $m \leq 2n - 4$ for a planar bipartite graph of n vertices and m edges.

Graph Coloring

Lemma 7.2.1 Every simple graph of the maximum degree Δ has a $(\Delta + 1)$ -coloring.

Theorem 7.2.2 Every simple graph of the maximum degree Δ which is not a complete graph has a Δ -coloring.

Every simple planar graph is 6-colorable

Every simple planar graph is 5-colorable

Theorem 7.3.1 *Let G be a simple graph with the maximum degree Δ . Then $\Delta \leq \chi'(G) \leq \Delta + 1$.*

Theorem 7.3.2 *Let G be a bipartite graph with the maximum degree Δ . Then $\chi'(G) = \Delta$.*

Here is a **clean summary** of the proof of König's line coloring theorem:

If G is a bipartite graph with maximum degree Δ , then

$$\chi'(G) = \Delta.$$

Summary of Proof

1. Lower bound

At a vertex with maximum degree Δ , all incident edges must have different colors, so:

$$\chi'(G) \geq \Delta$$

2. Goal

We need to show:

$$\chi'(G) \leq \Delta$$

i.e., we can color all edges using only Δ colors.

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3. Strategy (Induction + recoloring)

- Remove one edge.
 - By induction, color the remaining graph with Δ colors.
 - Try to insert the removed edge back.
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4. Key idea (free colors + alternating paths)

For the removed edge uv :

- Each endpoint has at least one unused color.
- If a common free color exists $\rightarrow$ assign it directly.
- Otherwise, use an **alternating color path (Kempe chain)** to swap colors and free a needed color.

Because the graph is bipartite, these swaps never create conflicts.

5. Conclusion

We can always extend a proper Δ -edge-coloring to the full graph.

So:

$$\chi'(G) \leq \Delta$$

Final result

Combining both bounds:

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If you want, I can reduce this further into a [↪ 3-line exam answer](#) or give a [↪ diagram-based intuition](#).

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For $n \geq 2$, $\chi'(Kn) = n$ if n is odd and $\chi'(Kn) = n - 1$ if n is even.